

§ 10.3 Polar Coordinates (극좌표)

Ex.8 까지

■ The Polar Coordinate System(극좌표계)

We choose a point in the plane that is called the **pole** (or origin) and is labeled O . Then we draw a ray (half-line) starting at O called the **polar axis**. This axis is usually drawn horizontally to the right and corresponds to the positive x -axis in Cartesian coordinates.

If P is any other point in the plane, let r be the distance from O to P and let θ be the angle (usually measured in radians) between the polar axis and the line OP as in Figure 1.

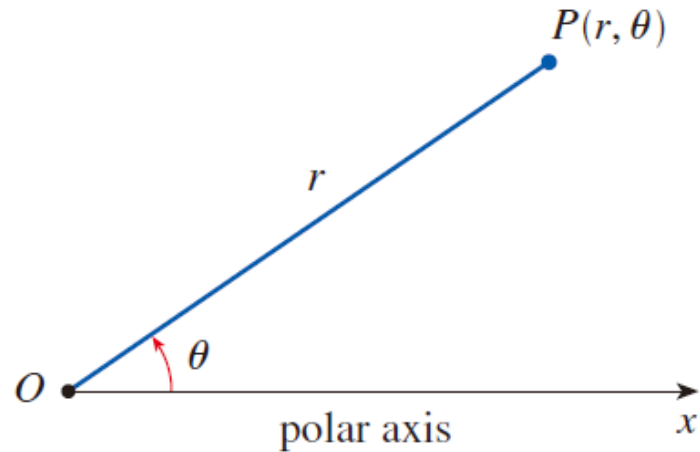


FIGURE 1

EXAMPLE 1 Plot the points whose polar coordinates are given.

- (a) $(1, 5\pi/4)$ (b) $(2, 3\pi)$ (c) $(2, -2\pi/3)$ (d) $(-3, 3\pi/4)$

Solution:

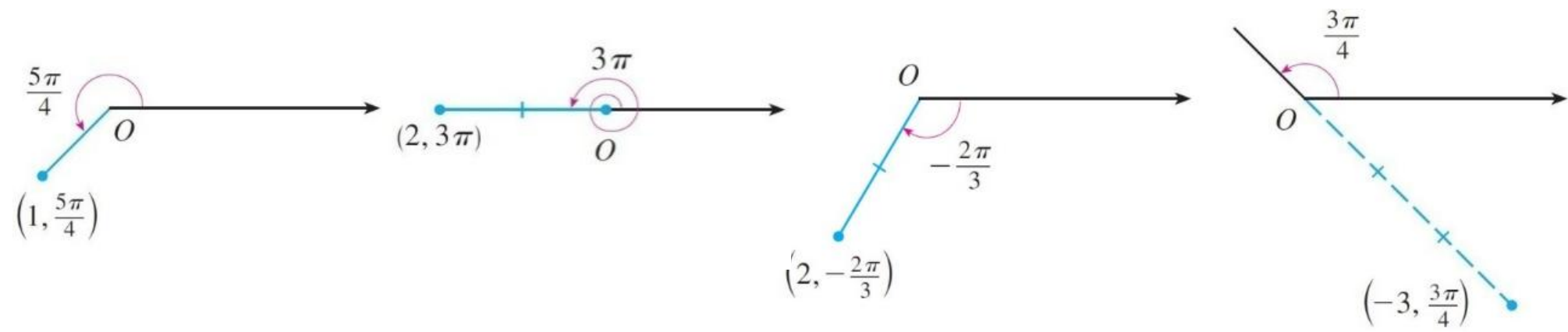


Figure 3

■ Relationship between Polar and Cartesian Coordinates

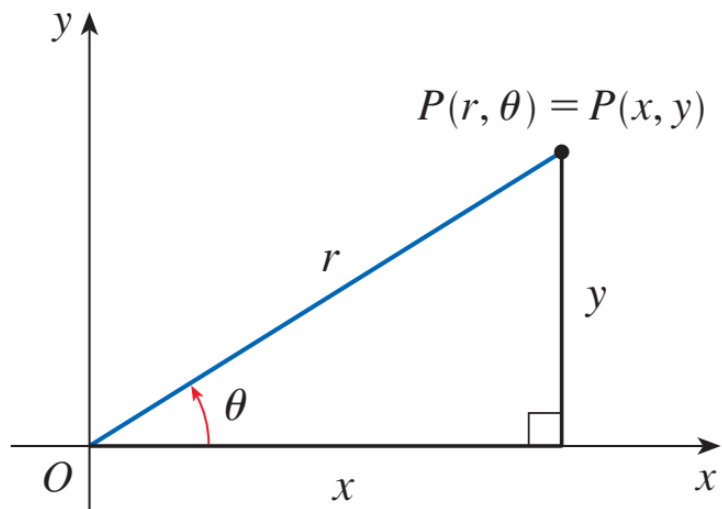


Figure 5

1

$$x = r \cos \theta \quad y = r \sin \theta$$

2

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x}$$

EXAMPLE 2 Convert the point $(2, \pi/3)$ from polar to Cartesian coordinates.

Solution:

EXAMPLE 3 Represent the point with Cartesian coordinates $(1, -1)$ in terms of polar coordinates.

Solution:

■ Polar Curves(극곡선)

The **graph of a polar equation** $r = f(\theta)$, or more generally $F(r, \theta) = 0$, consists of all points P that have at least one polar representation (r, θ) whose coordinates satisfy the equation.

EXAMPLE 4 What curve is represented by the polar equation $r = 2$?

Solution:

EXAMPLE 5 Sketch the polar curve $\theta = 1$.

Solution:

EXAMPLE 6

- (a) Sketch the curve with polar equation $r = 2 \cos \theta$.
- (b) Find a Cartesian equation for this curve.

Solution:

EXAMPLE 7 Sketch the curve $r = 1 + \sin \theta$.

Solution:

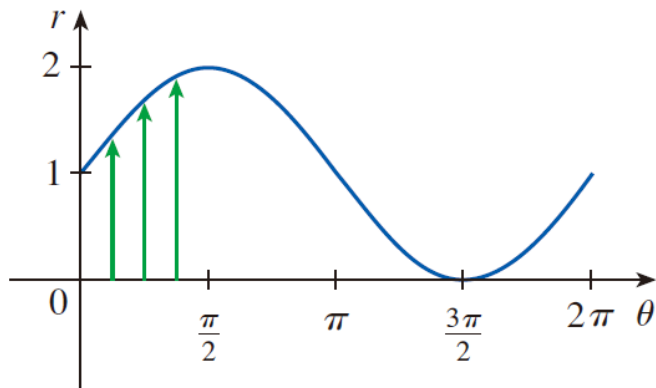


FIGURE 10

$r = 1 + \sin \theta$ in Cartesian
coordinates, $0 \leq \theta \leq 2\pi$

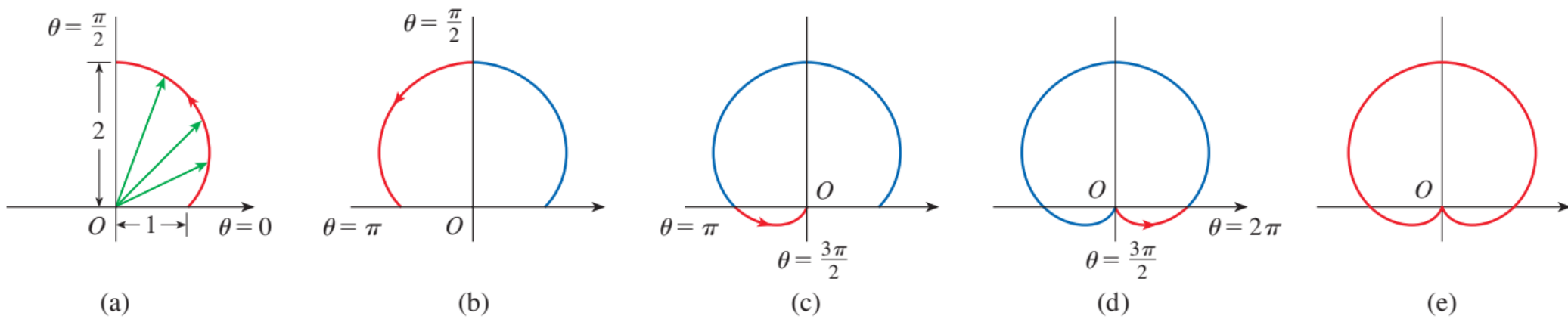


FIGURE 11 Stages in sketching the cardioid $r = 1 + \sin \theta$

EXAMPLE 8 Sketch the curve $r = \cos 2\theta$.

Solution:

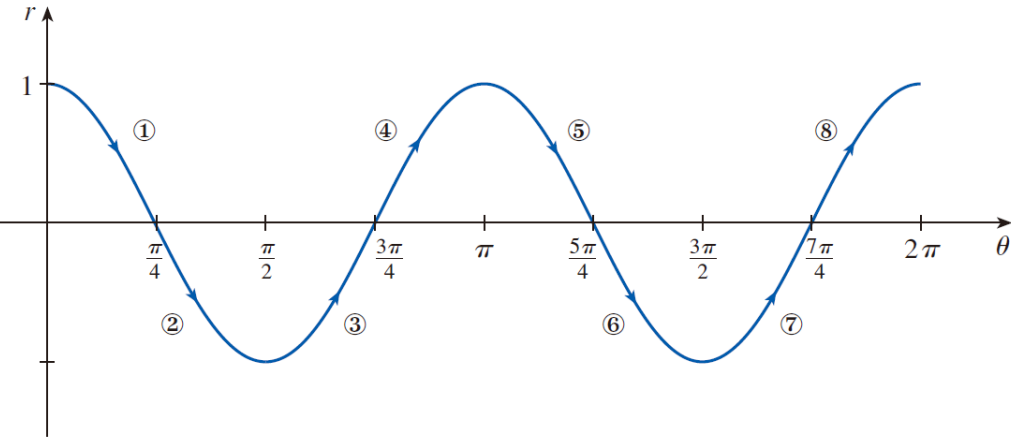


FIGURE 12
 $r = \cos 2\theta$ in Cartesian coordinates

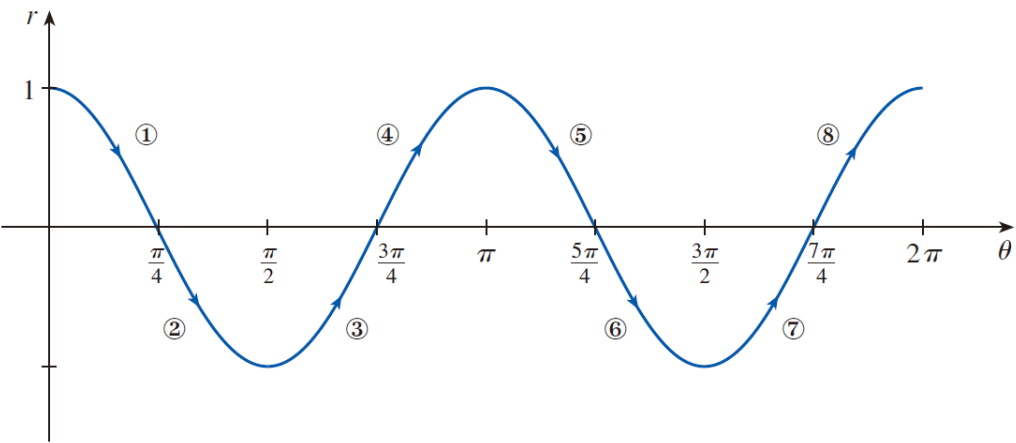


FIGURE 12
 $r = \cos 2\theta$ in Cartesian coordinates

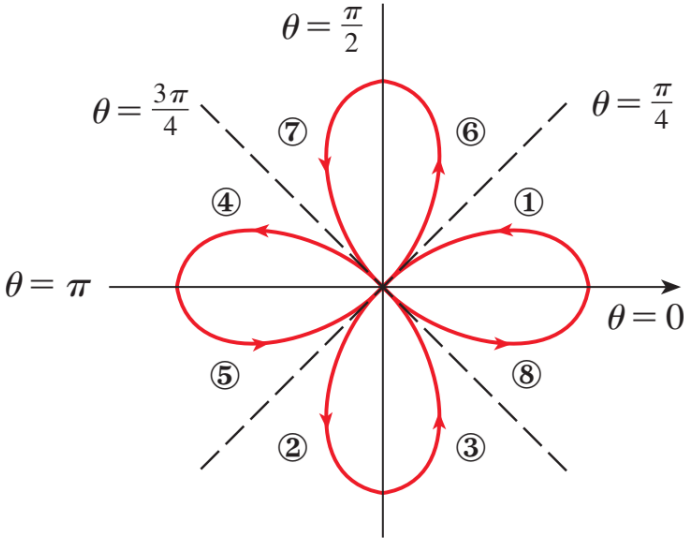


FIGURE 13
Four-leaved rose $r = \cos 2\theta$

